

MongoDB

NOSQL DB

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Summary

Introduction



mongo DB

- MongoDB is a document-oriented NoSQL database
- Stores data in JSON-like BSON documents rather than traditional rows and tables
- Documents are organized into collections
- Flexible schema allows documents in the same collection to have different fields
- Designed for scalability, flexibility, and high-volume data

History

2007

Dwight Merriman,
Eliot Horowitz, and
Kevin Ryan create
10gen

2013

10gen renamed itself
to MongoDB

Today

One of the fastest-
growing public
database companies
in the world

2009

Launched MongoDB

2017

MongoDB became a
publicly traded
company

SQL

```
1 CREATE TABLE students (  
2     id INT PRIMARY KEY,  
3     name VARCHAR(100),  
4     major VARCHAR(100)  
5 );  
6  
7 CREATE TABLE skills (  
8     id INT PRIMARY KEY,  
9     name VARCHAR(100)  
10 );  
11  
12 CREATE TABLE student_skills (  
13     student_id INT,  
14     skill_id INT,  
15     PRIMARY KEY (student_id, skill_id),  
16     FOREIGN KEY (student_id) REFERENCES students(id),  
17     FOREIGN KEY (skill_id) REFERENCES skills(id)  
18 );
```

```
20 INSERT INTO students VALUES  
21 (1, 'Dylan', 'Software Engineering');  
22  
23 INSERT INTO skills VALUES  
24 (1, 'Java'),  
25 (2, 'Python'),  
26 (3, 'MongoDB');  
27  
28 INSERT INTO student_skills VALUES  
29 (1, 1),  
30 (1, 2),  
31 (1, 3);  
32
```

MongoDB (NoSQL)

```
1  {  
2    "id": 1,  
3    "name": "Dylan",  
4    "major": "Software Engineering",  
5    "skills": ["Java", "Python", "MongoDB"]  
6  }
```

MongoDB (NoSQL)...Uh oh



```
1 {  
2   "id": 1,  
3   "name": "Dylan",  
4   "major": "Software Engineering"  
5 }  
1 {  
2   "id": 1,  
3   "name": "Dylan",  
4   "major": "Software Engineering",  
5   "skills": ["Java", "Python", "MongoDB"]  
6 }  
1 {  
2   "id": 1,  
3   "name": "Dylan",  
4   "major": "Software Engineering",  
5   "skills": ["Java", "Python", "MongoDB"]  
6 }  
1 {  
2   "id": 1,  
3   "name": "Dylan",  
4   "major": "Software Engineering",  
5   "skills": ["Java", "Python", "MongoDB"]  
6 }  
1 {  
2   "id": 1,  
3   "name": "Dylan",  
4   "major": "Software Engineering",  
5   "skills": ["Java", "Python", "MongoDB"]  
6 }  
1 {  
2   "id": 1,  
3   "name": "Dylan",  
4   "major": "Software Engineering",  
5   "skills": ["Java", "Python", "MongoDB"]  
6 }
```

MongoDB (NoSQL) with Schemas

```
1  const mongoose = require('mongoose');
2
3  // 1. Define the Schema
4  const studentSchema = new mongoose.Schema({
5    id: {
6      type: Number,
7      required: true,
8      unique: true
9    },
10   name: {
11     type: String,
12     required: true
13   },
14   major: {
15     type: String,
16     required: true
17   },
18   skills: {
19     type: [String]
20   }
21 });
22
23 // Mongoose automatically pluralizes 'Student' to target the 'students' collection in MongoDB
24 const Student = mongoose.model('Student', studentSchema);
25
26 module.exports = Student;
```

Demo

- CRUD = Create, Read, Update, Delete



Professional Usage



Learn how job-matching site Indeed reduced its data storage, transfer, and compute costs by 27% on MongoDB Atlas.



Adobe uses MongoDB, an AI-ready data platform, to power its generative and agentic tools.



Learn how Coinbase leveraged MongoDB Atlas to decrease scaling time by more than 60%.



To power and scale its natural language queries, Okta implemented MongoDB Atlas Vector Search to automate...



Learn how multinational bank, Wells Fargo, leveraged MongoDB to modernize its card payments platform to handle 7M+...



TOYOTA
Financial Services

Toyota Financial Services uses MongoDB Atlas to make better decisions and build better products.

— Why Should You Use It?

Flexible

*Changing or Evolving
Data*

Scalable

Growing Data & Users

Fast

*High-Volume
Applications*

Document-Based

*Data Naturally Fits
Together*

Developer-Friendly

Easy to Build & Iterate

Distributed

*Multiple Servers or
Locations*

— Summary

- ① MongoDB is NoSQL, using a document-oriented database with flexible BSON documents
- ② Shemas can easily change
- ③ Highly scalable (horizontally through sharding), meant for large amounts of data and traffic
- ④ Developer-friendly, documents map to data / objects very easily

The End

ANY QUESTIONS?

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